amazon rme

AMAZON BASE BUILDING MAINTENANCE STANDARD

GLOBAL CENTRAL RME STANDARDS & COMPLIANCE

Task Title	FIRE SUPPRESSION SYSTEM, FIRE PUMPS, FIRE PUMP 01 - 10
APM Object	FIRE.SUP.PUMP.AS01
APM EC	AMZ.FIRE.PUMP.AS
BBM Handbook Title	Fire Pumps - Diesel and Electric
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure These tasks are based on industry guidance, SME knowledge and HIE learning. Please ensure vendor specific requirements are followed. To suggest improvements please raise a SIM SIM - Link to SIM template

Last Update: (Version)	Jun-24	**Geo Applied:**	Intl APM Instance	**F-Gas**	Not Identified	**Activies may be covered by No-Go restrictions:**	TBC	**Special Certificaion Required:**	TBC	**Delivered By:**	Carried out by trained on site technician certified in the electrical field	**OCB:**	BBM
Evidence for CT:	YES	**Pricing template Position:**	10	**Related CT Areas:**	CT.FIRE.PUMP			**Assets included in this PM:**	PUMP FUEL TANK - DIESEL ENGINE PUMP - AC MOTOR PUMP - PUMPS CONTROL PANEL	**Schedule With:**	FIRE.SUP.PUMP.JP01		

Description of System / Technology (According to BBM Handbook)

Diesel fire pumps, also known as booster pumps, are used to boost water pressure from sources that do not provide adequate water pressure for the sprinklers to operate to a given specification. A fire pump can be connected to a city water main, tank or reservoir/pond used for firefighting purposes.

Additional Information

All sites with the above listed Base Building Asset should conduct routine maintenance and inspections according to this standard PM package. The Amazon maintenance standard supports compliance, reduces safety risks,improves asset reliability & efficiency.
Some maintenances captures several assets within one PM package -> Ensure that corrective/reactive maintenance for assets where proactive maintenance is contained under another asset is created against the original asset.

Important
Forecast PM together with PM of the control panel (fire suppression controls)

Task Index	Component	Tasks	Tasks Nested for frequencies > weekly									
			Daily	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	36 Monthly	48 Monthly	60 Monthly
1	General check	Is the room properly ventilated						PM				
2	General check	Are the components clean - free from debris & leakage						PM				
3	General check	Check of the equipment for damage & missing components						PM				
4	General check	Check fuel tanks for damage & leakage						PM				
5	General check	Check collection tray for deposits FPM - identify the root cause in case of deposit						PM				
6	Documentation	Are the circuit diagrams and relevant technical documentation available and complete						PM				
7	Documentation	Are there manual entries in the drawings or technical documentation that indicate changes to the installation						PM				
8	Documentation	Verify that system schematics are properly mounted on the walls and clearly visible for easy reference						PM				
9	Safety	Are all relevant safety notices visible and free from damage						PM				
10	Safety	Is appropriate fire-fighting equipment available in the rooms						PM				
11	Safety	Are disconnecting devices lockable in the safe position (LOTO)						PM				
12	Safety	Test all safety relevant switches						PM				
13	Battery & Charger	Check if the battery charging device is working correctly (Correct charge/float voltage)		PM								
14	Battery & Charger	Measure the voltage of each battery (Off-charge, using test button or non-intrusive method where possible)						PM				
15	Battery & Charger	Check the batteries for leakage		PM								
16	Battery & Charger	Check the fluid level of the batteries and refill if required						PM				
17	Battery & Charger	Schedule a battery replacement immediately upon detecting any deficiency or performance issue		PM								
18	Fire Pump Fuel Tank	Are all components labeled correctly						PM				
19	Fire Pump Fuel Tank	Visually inspect all fuel system components for leakage						PM				
20	Fire Pump Fuel Tank	Carry out a functional cleaning of the fuel system						PM				
21	Fire Pump Fuel Tank	Are disconnecting devices lockable in the safe position (LOTO)						PM				
22	Fire Pump Fuel Tank	Inspect fuel pumps for damage, corrosion and unusual noise (if the fuel pump is installed)						PM				
23	Fire Pump Fuel Tank	Ensure the fuel pump, if installed, is securely fixed and properly aligned						PM				
24	Fire Pump Fuel Tank	Check if the fuel pump, if installed, is functioning correctly						PM				
25	Fire Pump Fuel Tank	Change fuel filter						PM				
26	Fire Pump Fuel Tank	Check valves for leakage, corrosion & damage						PM				
27	Fire Pump Fuel Tank	Inspect all valves, including automatic and safety valves, to confirm proper operation and functionality						PM				
28	Fire Pump Fuel Tank	Check fuel tank for leakage corrosion & damage						PM				
29	Fire Pump Fuel Tank	Check position/limit value transmitter and limit position switch for correct function						PM				
30	Fire Pump Fuel Tank	Ensure that fuel tank is always full to enable supression in case of a fire		PM				PM				
31	Fire Pump Diesel Motor	Check the Diesel pumps engine for oil leaks		PM								
32	Fire Pump Diesel Motor	Check oil level, with engine not running		PM								
33	Fire Pump Diesel Motor	Check for Diesel pumps engine coolant leaks		PM								
34	Fire Pump Diesel Motor	Check engine block heater temperature [the temperature should be at least 120 F (50 C)]		PM								
35	Fire Pump Diesel Motor	Assess whether the air filter requires cleaning or replacement based on its condition		PM								

36	Fire Pump Diesel Motor	Check if oil needs replacement (based on date or operational hours)		PM								
37	Fire Pump Diesel Motor	For locations with water tanks – Verify that the isolation valve on the tank return line is open before pump starts		PM								
38	Fire Pump Diesel Motor	Start pump in auto React appropriately if there were any problems when pump started		PM								
39	Fire Pump Diesel Motor	Check oil pressure (Typical: 40-70 psi (2.8 – 4.8 bars)		PM								
40	Fire Pump Diesel Motor	Check the sight glass to make sure water is flowing through to the heat exchanger		PM								
41	Fire Pump Diesel Motor	Check the temperature gauge to make sure that engine is running at 160-200 F (70-93 C. Stop pump immediately if heat reaches 200F (93C)		PM								
42	Fire Pump Diesel Motor	Check exhaust for excessive smoke (white, blue or black) If smoke is observed stop engine immediately!!		PM								
43	Fire Pump Diesel Motor	Record oil pressure while diesel is running		PM								
44	Fire Pump Diesel Motor	Record operational hour meter reading		PM								
45	Fire Pump Diesel Motor	Check for water dripping from packing at both ends (should be no more than one drip a second when pump is NOT in use).		PM								
46	Fire Pump Diesel Motor	Is the level of the cooling fluid ok -> refill if required						PM				
47	Fire Pump Diesel Motor	Check for proper corrosion and freezing protection of the cooling fluid						PM				
48	Fire Pump Diesel Motor	Check hoses & pipes for damage & wear						PM				
49	Fire Pump Diesel Motor	Check connections at hoses & pipes for tightness						PM				
50	Fire Pump Diesel Motor	Check the engine starter and alternator for correct function						PM				
51	Fire Pump Diesel Motor	Check all value transmitters and display elements for damage and correct function						PM				
52	Fire Pump Diesel Motor	Check Engine clutch for correct function						PM				
53	Fire Pump Diesel Motor	Are there any unusual noises during start, operation or stop of the engine						PM				
54	Fire Pump Diesel Motor	Grease all possible greasing points						PM				
55	Fire Pump Diesel Motor	Check that the screws and nuts are tight						PM				
56	Fire Pump Diesel Motor	Check engine block for leakage						PM				
57	Fire Pump Diesel Motor	Verify that the exhaust system is securely sealed and free from any damage or defects						PM				
58	Fire Pump Diesel Motor	Exchange jacket heater hose							PM			
59	Fire Pump Diesel Motor	Is the jacket heater thremostat functioning and the thresholds are set correctly						PM	PM			
60	Fire Pump Diesel Motor	Replace coolant and ensure the correctmix according to manufacturer specifications							PM			
61	Fire Pump Diesel Motor	Verify correct coolant protection level (environment & manufacturer specification)					PM		PM			
62	Fire Pump Diesel Motor	Replace collant hoses							PM			
63	Fire Pump Diesel Motor	Replace coolant water pump							PM			
64	Fire Pump Diesel Motor	Replace batteries							PM			
65	Fire Pump Diesel Motor	Replace thermostat							PM			
66	Fire Pump Diesel Motor	Replace belts							PM			
67	Fire Pump Diesel Motor	Verify that hoses are clear of contact with hot surfaces						PM	PM			
68	Fire Pump Diesel Motor	Record operational hours in the notes						PM	PM			
69	Fire Pump Electrical Motor	Check condition of anti vibration mounts						PM	PM			
70	Fire Pump Electrical Motor	Measure the electrical current flow and compare it with the design specification						PM	PM			
71	Fire Pump Electrical Motor	Torque check of motor cable terminations according to OEM to torque requirements						PM	PM			
72	Fire Pump Electrical Motor	Check if the motor generates any unusual noise or vibration during operation						PM	PM			
73	Fire Pump Electrical Motor	Check for overtemperature of the motor						PM	PM			
74	Fire Pump Electrical Motor	Clean the cooling fins and the air inlet screen						PM	PM			
75	Fire Pump Electrical Motor	Lubricate if applicable						PM	PM			
76	Fire Pump	Check the pump for unusual noise and lubricate bearings						PM	PM			
77	Fire Pump	Check pump seals for exessive leakage						PM	PM			
78	Field devices	Check the automatic & correct function of all air in- & outlet flaps						PM	PM			
79	Emergency check	Simulate a power outage of the regular power supply of the control panel						PM	PM			
80	Emergency check	Check if sprinkler pumps start & operation is possible						PM	PM			
81	Pumps Control	Are disconnecting devices lockable in the safe position (LOTO)						PM	PM			
82	Pumps Control	Conduct thermographic scans and create FMP to identify and address the root cause of any over-temperature issues						PM	PM			
83	Pumps Control	Add a detailed note for all over-temperature measurements -> list the location and temperature						PM	PM			
84	Pumps Control	Upload the thermographic scans in case of overtemperature						PM	PM			
85	Pumps Control	Is the panel/cabinet clean - free from debris						PM	PM			
86	Pumps Control	Change the air filter of the electrical panel/cabinet						PM	PM			
87	Pumps Control	Is the cooling fan operating appropriately and doesn`t make any unusual noise						PM	PM			
88	Pumps Control	Visual check of the panel outside & inside for damage & missing components						PM	PM			
89	Pumps Control	Are there any alarms indicated -> if yes identify the root cause						PM	PM			
90	Pumps Control	Are any circuit breakers tripped						PM	PM			
91	Pumps Control	Visual check of cable connections such as pinched insulation in the contact area						PM	PM			
92	Pumps Control	Visual check for the protection against direct contact with live parts						PM	PM			
93	Pumps Control	Were any obvious changes made (e.g. rewiring, hanging cables)						PM	PM			
94	Pumps Control	Are all cables properly routed						PM	PM			
95	Pumps Control	Are all components properly labeled or marked						PM	PM			
96	Pumps Control	Test all safety relevant switches						PM	PM			

97	Pumps Control	Are all components properly attached						PM	PM				
98	Pumps Control	Ensure all cable connections are correctly torqued according to manufacturers specification						PM	PM				
99	Pumps Control	Are all metal parts connected to the protection earth (ground)						PM	PM				
100	Pumps Control	Is the ground connection to the building ground as designed?						PM	PM				
101	Pumps Control	Check if the panel is connected to the emergency power supply correctly (if applicable)						PM	PM				
102	Pumps Control	Carry out functional cleaning of relevant components						PM	PM				
103	Functional cleaning	Clean the air intakes on the engine, generator and main air intake grilles on the outer walls						PM	PM				
104	Functional cleaning	Carry out a functional cleaning of all equipment if required						PM	PM				
105	Final Check	Check the O&M manual & local requirements (e.g. fire protection concept & building permit) and carry out additionally required tasks						PM	PM				

Change Log

July 2023 - First issue in new format. Simplification of task scope, alignment with CT.
June 2024 - Changed frequency, changed the whole PM and added tasks for electrical pump motor, added tasks from 10 - FIRE SUPRESSION CONTROL PANELS. Removed task regarding CE certification.